

Homework 10 Solutions

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Problem 21.2

Proof. $1 + 2 + 3 + \dots + n = \frac{1}{2}(n)(n+1)$ for all positive n .

1. Assume that there is some smallest value x for which the statement is false.
2. Observe that $1 = \frac{1}{2}(1)(1+1)$, and so $x > 1$.
3. Since x is the smallest value for which the statement does not hold, it must hold for $x-1$.
4. Then $1 + 2 + 3 + \dots + (x-1) = \frac{1}{2}(x-1)(x-1+1)$.
5. Then $1 + 2 + 3 + \dots + (x-1) + x = \frac{1}{2}(x-1)(x-1+1) + x$.
6. $\frac{1}{2}(x-1)(x-1+1) + x = \frac{1}{2}(x-1)(x) + x = \frac{1}{2}(x^2 - x + 2x) = \frac{1}{2}(x)(x+1)$.
7. So $1 + 2 + 3 + \dots + (x-1) + x = \frac{1}{2}(x)(x+1)$, contradicting the definition of $x \Rightarrow \Leftarrow$. □

Problem 21.3

Proof. $n < 2^n$ for all natural numbers.

1. Assume that there is some smallest value x for which the statement is false.
2. Observe that $0 < 2^0 = 1$, and so $x > 0$.
3. Since x is the smallest value for which the statement does not hold, it must hold for $x-1$.
4. Then $(x-1) < 2^{x-1}$.
5. Then $(x-1)(x) < 2^{x-1}x = 2^x$.
6. Since x is always positive, then $x < (x-1)(x) < 2^x$.
7. So $x < 2^x$, contradicting the definition of $x \Rightarrow \Leftarrow$. □

Problem 21.4

Proof. $n! \leq n^n$ for all natural numbers.

1. Assume that there is some smallest value x for which the statement is false.
2. Observe that $1 = 0! \leq 0^0 = 1$, and so $x > 0$.
3. Since x is the smallest value for which the statement does not hold, it must hold for $x-1$.
4. Then $(x-1)! \leq (x-1)^{x-1}$.
5. Then $(x-1)!(x) \leq (x-1)^{x-1}x$.
6. Since x is always positive, then $(x-1)!(x) = x! \leq (x-1)^{x-1}x < x^{x-1}x = x^x$.
7. So $x! < x^x$, contradicting the definition of $x \Rightarrow \Leftarrow$. □

Problem 22.4

a) *Proof.* $1 + 4 + 7 + \dots + (3n-2) = \frac{n(3n-1)}{2}$.

1. Base case $n = 1$: $1 = \frac{1(2)}{2} = \frac{1(3 \cdot 1 - 1)}{2}$.

2. Assume that when $n = k$, $1 + 4 + 7 + \dots + (3k-2) = \frac{k(3k-1)}{2}$.

3. Then $1 + 4 + 7 + \dots + (3k-2) + [3(k+1)-2] = \frac{k(3k-1)}{2} + [3(k+1)-2]$.

4. Then $\frac{k(3k-1)}{2} + [3(k+1)-2] = \frac{3k^2-k}{2} + (3k+1) = \frac{3k^2-k+2(3k+1)}{2} = \frac{3k^2-k+6k+2}{2} = \frac{3k^2+5k+2}{2} = \frac{(k+1)(3k+2)}{2} = \frac{(k+1)(3(k+1)-1)}{2}$.

5. Therefore $1 + 4 + 7 + \dots + (3k-2) + [3(k+1)-2] = \frac{(k+1)(3(k+1)-1)}{2}$, showing that it holds when $n = k+1$. □

b) *Proof.* $1^3 + 2^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$.

1. Base case $n = 1$: $1 = 1^3 = \frac{1^2(1+1)^2}{4} = 1$.

2. Assume that when $n = k$, $1^3 + 2^3 + \dots + k^3 = \frac{k^2(k+1)^2}{4}$.

3. Then $1^3 + 2^3 + \dots + k^3 + (k+1)^3 = \frac{k^2(k+1)^2}{4} + (k+1)^3$.

4. So $\frac{k^2(k+1)^2}{4} + (k+1)^3 = \frac{k^2(k+1)^2 + 4(k+1)^3}{4} = \frac{(k+1)^2(k^2 + 4k + 4)}{4} = \frac{(k+1)^2(k+2)^2}{4}$.

5. Therefore $1^3 + 2^3 + \dots + k^3 + (k+1)^3 = \frac{(k+1)^2(k+2)^2}{4}$, showing that it holds when $n = k+1$. $\square$

c) *Proof.* $9 + 9 \times 10 + 9 \times 100 + \dots + 9 \times 10^{n-1} = 10^n - 1$.

1. Base case $n = 1$: $9 = 10 - 1 = 10^1 - 1$.

2. Assume that when $n = k$, $9 + 9 \times 10 + 9 \times 100 + \dots + 9 \times 10^{k-1} = 10^k - 1$.

3. Then $9 + 9 \times 10 + 9 \times 100 + \dots + 9 \times 10^{k-1} + 9 \times 10^k = 10^k - 1 + 9 \times 10^k = 10 \times 10^k - 1 = 10^{k+1} - 1$.

4. Therefore $9 + 9 \times 10 + 9 \times 100 + \dots + 9 \times 10^{k-1} + 9 \times 10^k = 10^{k+1} - 1$, showing that it holds when $n = k+1$. $\square$

d) *Proof.* $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{n(n+1)} = 1 - \frac{1}{n+1}$.

1. Base case $n = 1$: $\frac{1}{2} = \frac{1}{1 \times 2} = 1 - \frac{1}{1+1} = \frac{1}{2}$.

2. Assume that when $n = k$, $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{k(k+1)} = 1 - \frac{1}{k+1}$.

3. Then $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} = 1 - \frac{1}{k+1} + \frac{1}{(k+1)(k+2)}$.

4. So $1 - \frac{1}{k+1} + \frac{1}{(k+1)(k+2)} = 1 + \frac{-k-2+1}{(k+1)(k+2)} = 1 + \frac{-(k+1)}{(k+1)(k+2)} = 1 - \frac{1}{(k+1)(k+2)}$.

5. Therefore $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} = 1 - \frac{1}{(k+1)(k+2)}$, showing that it holds when $n = k+1$. $\square$

Problem 22.5

a) *Proof.* $2^n \leq 2^{n+1} - 2^{n-1} - 1$.

1. Base case $n = 1$: $2 = 2^1 \leq 2^2 - 2^0 - 1 = 2$.

2. Assume that when $n = k$, $2^k \leq 2^{k+1} - 2^{k-1} - 1$.

3. Then $2^{k+1} = 2 \times 2^k \leq 2(2^{k+1} - 2^{k-1} - 1) = (2^{k+2} - 2^k - 1) - 1$.

4. Therefore $2^{k+1} \leq 2^{k+2} - 2^k - 1$, showing that it holds when $n = k+1$. $\square$

c) *Proof.* $1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^n} \geq 1 + \frac{n}{2}$.

1. This one seems to be a typo of some sort.

2. Let $n = 2$, then $1 + \frac{1}{2} + \frac{1}{4} \geq 1 + \frac{2}{2}$ is false. $\square$

e) *Proof.* $n! \leq n^n$.

1. Base case $n = 1$: $1 = 1! \leq 1^1 = 1$.

2. Assume that when $n = k$, $k! \leq k^k$.

3. Then $(k+1)! = (k+1)k! \leq (k+1)k^k < (k+1)(k+1)^k = (k+1)^{k+1}$.

4. Therefore $(k+1)! \leq (k+1)^{k+1}$, showing that it holds when $n = k+1$. $\square$

Problem 22.11

Proof.

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

1. Base case $n = 1$:

$$x+y = (x+y)^1 = \sum_{k=0}^1 \binom{1}{k} x^{1-k} y^k = \binom{1}{0} x^1 y^0 + \binom{1}{1} x^0 y^1 = x+y.$$

2. Assume that when $n = j$,

$$(x+y)^j = \sum_{k=0}^j \binom{j}{k} x^{j-k} y^k.$$

3. Multiply both sides by $(x + y)$

$$(x + y)^{j+1} = (x + y)^j(x + y) = (x + y) \sum_{k=0}^j \binom{j}{k} x^{j-k} y^k.$$

4. Distribute the $(x + y)$

$$= \sum_{k=0}^j \binom{j}{k} x^{j+1-k} y^k + \sum_{k=0}^j \binom{j}{k} x^{j-k} y^{k+1}.$$

5. Group all k 's in the right summation as $(k + 1)$

$$= \sum_{k=0}^j \binom{j}{k} x^{j+1-k} y^k + \sum_{k=0}^j \binom{j}{(k+1)-1} x^{j-(k+1)+1} y^{(k+1)}.$$

6. Adjust the right summation to start at 1

$$= \sum_{k=0}^j \binom{j}{k} x^{j+1-k} y^k + \sum_{k=1}^{j+1} \binom{j}{(k)-1} x^{j-(k)+1} y^{(k)}.$$

7. Compute $k = 0$ in the left summation and $k = j + 1$ in the right

$$= \binom{j}{0} x^{j+1-0} y^0 + \sum_{k=1}^j \binom{j}{k} x^{j+1-k} y^k + \binom{j}{j+1-1} x^{j-j-1+1} y^{j+1} + \sum_{k=1}^j \binom{j}{k-1} x^{j+1-k} y^k.$$

8. Combine the two summations and simplify the two constants

$$= \binom{j}{0} x^{j+1} y^0 + \binom{j}{j} x^0 y^{j+1} + \sum_{k=1}^j [\binom{j}{k-1} + \binom{j}{k}] x^{j+1-k} y^k.$$

9. Apply Pascal's Identity and manipulate constants to use $j + 1$

$$= \binom{j+1}{0} x^{j+1} y^0 + \binom{j+1}{j+1} x^0 y^{j+1} + \sum_{k=1}^j \binom{j+1}{k} x^{j+1-k} y^k.$$

10. Account for the two constants in the summation ($k = 0$ and $k = j + 1$)

$$(x + y)^{j+1} = \sum_{k=0}^{j+1} \binom{j+1}{k} x^{j+1-k} y^k.$$

11. Therefore it holds when $n = j + 1$. □